PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Machine Learning-Based Detection of Phishing Websites Using URL, Domain, and Webpage Features

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled "Machine Learning-Based Detection of Phishing Websites Using URL, Domain, and Webpage Features" in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of our faculty mentor. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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We also acknowledge the UCI Machine Learning Repository for making the Phishing Websites Dataset publicly available, which served as the backbone of our experimental work. Without this open dataset, our comparative evaluation of machine learning classifiers would not have been possible.

Finally, we thank our family and friends for their patience and moral support throughout the duration of this project.

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ABSTRACT

Phishing attacks represent one of the most serious cybersecurity threats worldwide. Attackers create fake websites that closely replicate trusted services such as banks, e-commerce platforms, and social media portals to trick users into surrendering their passwords and financial details. Traditional blacklist-based detection systems are inherently reactive and fail against newly created phishing domains that are only active for 24 to 48 hours.

This project presents a machine learning-based framework for real-time phishing website detection using 30 features extracted from URL structure, domain registration metadata, and HTML/JavaScript content. Four supervised classification algorithms were implemented and rigorously evaluated: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). All models were trained and tested on the UCI Phishing Websites Dataset comprising 11,055 labeled samples under identical experimental conditions to ensure a fair and reproducible comparison.

The Random Forest ensemble classifier achieved the best overall performance with 97% accuracy, 96% precision, 96% recall, 96% F1-score, and an Area Under the ROC Curve (AUC) of 0.99. These results confirm that ensemble-based machine learning methods significantly outperform individual classifiers for phishing detection. The framework does not require real-time page rendering or external blacklist queries, making it computationally efficient and suitable for deployment as a real-time browser extension or network-level security filter.

Keywords: Phishing Detection, Machine Learning, Random Forest, Feature Extraction, Cybersecurity, URL Analysis, Ensemble Classifier, Support Vector Machine

1. INTRODUCTION

1.1 Background

Phishing is a type of social engineering attack in which cybercriminals construct fraudulent websites that visually and structurally replicate legitimate online services. Users who visit these fake pages are tricked into entering sensitive information — passwords, credit card numbers, bank account details — which is immediately captured by the attacker. According to the Anti-Phishing Working Group (APWG), over 4.7 million phishing attacks were recorded in 2022, representing a 150% increase compared to 2019 figures. The financial losses caused by phishing attacks amount to billions of dollars annually, affecting individuals, businesses, and government organizations alike.

The most widely used method to combat phishing is blacklisting — maintaining regularly updated databases of known malicious URLs, such as those managed by PhishTank and Google Safe Browsing. While effective against previously identified threats, blacklisting suffers from a critical weakness: phishing websites are typically active for only 24 to 48 hours before being taken down and replaced with new domains. By the time a new phishing site is discovered and added to a blacklist, it may have already victimized thousands of users.

1.2 Problem Statement

The core challenge this project addresses is the automated binary classification of websites as either phishing or legitimate — without relying on blacklists — using only intrinsic features derived from the URL structure, domain registration metadata, and HTML/JavaScript content of the webpage. The goal is to develop a machine learning model that can accurately detect previously unseen phishing websites in real time, minimizing false negatives (missed phishing sites) as the primary security objective.

1.3 Objectives

The specific objectives of this project are:

O1. To extract and systematically analyze 30 discriminative features from URL strings, WHOIS/DNS domain records, and HTML/JavaScript content of websites.

O2. To implement and train four supervised machine learning classifiers — Logistic Regression, Decision Tree, Random Forest, and SVM — under identical experimental conditions.

O3. To rigorously evaluate and compare classifier performance using accuracy, precision, recall, F1-score, confusion matrix, and AUC-ROC metrics.

O4. To identify the most discriminative features for phishing detection using Random Forest feature importance analysis.

O5. To develop an interactive web-based demonstration tool for real-time phishing URL analysis.

1.4 Scope of the Project

This project covers the complete machine learning pipeline from feature extraction to model evaluation. It focuses on classical supervised learning algorithms and does not include deep learning approaches. The dataset used is the publicly available UCI Phishing Websites Dataset. The project also includes an interactive HTML-based demo application that allows real-time URL analysis without requiring any server infrastructure.

1.5 Organization of the Report

Chapter 2 describes the methodology including dataset description, preprocessing, and model training. Chapter 3 covers tools and technologies used. Chapter 4 details the implementation. Chapter 5 presents major findings and results. Chapter 6 concludes the report with future work directions. References and appendices follow.

2. METHODOLOGY

2.1 Dataset Description

This project uses the UCI Phishing Websites Dataset, a widely used benchmark in phishing detection research introduced by Mohammad et al. (2014). Table 2 summarizes the dataset properties.

Table 2: Dataset Overview — UCI Phishing Websites Dataset

Property	Details
Total Samples	11,055 websites
Phishing Samples	5,715 (51.7%)
Legitimate Samples	5,340 (48.3%)
Number of Features	30
Feature Encoding	-1 (phishing), 0 (suspicious), +1 (legitimate)
Missing Values	None
Source	UCI Machine Learning Repository
Reference	Mohammad et al., Neural Computing and Applications, 2014

2.2 Feature Engineering

Each website in the dataset is described by 30 features organized into three categories: URL-based features (extracted from the URL string), Domain-based features (from WHOIS and DNS records), and HTML/JavaScript features (from the webpage source code). Table 1 presents the 10 most discriminative features.

Table 1: Top 10 Key Features Used for Classification

URL Length	URL	Length > 54 chars = suspicious; > 75 = phishing
Having IP Address	URL	IP address in URL hides true destination
@ Symbol in URL	URL	Browser ignores content before @ symbol
HTTPS Token in Domain	URL	"https-paypal.com" style tricks users
Subdomain Count	URL	More than 3 dots in URL = phishing indicator

Age of Domain	Domain	Domains < 6 months old = likely phishing
DNS Record Available	Domain	No DNS record = strong phishing signal
Web Traffic Rank	Domain	Low Alexa rank indicates legitimate site
IFrame Usage	HTML/JS	Hidden IFrames inject malicious content
Right-Click Disabled	HTML/JS	Prevents users from viewing page source

2.3 Data Preprocessing

Since all 30 features are pre-encoded as integers within {-1, 0, +1} and no missing values are present, minimal preprocessing was required. For tree-based classifiers (Decision Tree, Random Forest), features were used directly. For distance-sensitive classifiers (Logistic Regression, SVM), StandardScaler normalization was applied to transform all features to zero-mean, unit-variance distributions for stable convergence.

2.4 Train-Test Split

The dataset was split into 80% training (8,844 samples) and 20% testing (2,211 samples) using stratified random sampling with a fixed random seed of 42. Stratification ensures that the original 51.7%/48.3% class ratio is preserved in both sets. Five-fold stratified cross-validation was applied during hyperparameter tuning.

3. TOOLS AND TECHNOLOGIES

3.1 Programming Language

Python 3.x was used as the primary programming language for data preprocessing, model training, evaluation, and visualization. Python was chosen for its extensive ecosystem of machine learning libraries and ease of implementation.

3.2 Machine Learning Libraries

Scikit-learn: Used for implementing all four classifiers (Logistic Regression, Decision Tree, Random Forest, SVM), StandardScaler normalization, train-test split, cross-validation, and evaluation metrics including confusion matrix, classification report, and ROC-AUC score.

NumPy: Used for numerical operations and array manipulations during feature processing and metric calculations.

Pandas: Used for loading, inspecting, and managing the UCI dataset in DataFrame format.

3.3 Visualization

Matplotlib: Used for generating all charts and figures including the confusion matrix heatmap, ROC curves, grouped bar charts for metrics comparison, and literature comparison graphs.

Seaborn: Used for enhanced statistical visualizations including the confusion matrix color map.

3.4 Development Environment

VS Code: Primary code editor used for writing Python scripts and the HTML demo application.

Jupyter Notebook: Used for iterative experimentation, model testing, and result visualization during development.

3.5 Dataset Source

UCI Machine Learning Repository: Source of the Phishing Websites Dataset (Mohammad et al., 2014) used for all experiments.

3.6 Demo Application

HTML/CSS/JavaScript: Used to build the interactive phishing detector web application (phishing_detector.html). The application runs entirely in the browser without any server or installation requirement, making it portable and easy to demonstrate.

4. IMPLEMENTATION

4.1 Classification Algorithms

Four supervised machine learning algorithms were implemented and evaluated:

Logistic Regression (LR): A linear probabilistic classifier trained with L2 regularization (C = 1.0) and the lbfgs solver with a maximum of 1,000 iterations. Used as the linear baseline model.

Decision Tree (DT): A non-parametric recursive partitioning classifier using the Gini impurity criterion for split selection, with maximum depth constrained to 20 nodes to prevent overfitting.

Random Forest (RF): An ensemble of 100 independently trained decision trees using bootstrap sampling and random feature subsampling (max_features = $\sqrt{30} \approx 5$ features per split). Final predictions are made by majority voting across all 100 trees.

Support Vector Machine (SVM): A maximum-margin classifier using the Radial Basis Function (RBF) kernel with C = 10 and γ = 'scale'. The regularization parameter C = 10 was selected via 5-fold cross-validation grid search.

Table 3: Hyperparameters of All Four Classifiers

Model	Key Hyperparameters	Notes
Logistic Regression	C=1.0, penalty=L2, solver=lbfgs, max_iter=1000	Linear baseline
Decision Tree	criterion=gini, max_depth=20	Depth limited to prevent overfitting
Random Forest	n_estimators=100, max_features=√30, bootstrap=True	Best performing model
SVM (RBF)	kernel=rbf, C=10, gamma=scale	C chosen via cross-validation

4.2 Evaluation Metrics

All models were evaluated on the same 2,211-sample held-out test set using the following metrics:

- Accuracy — overall proportion of correct predictions
- Precision — fraction of predicted phishing sites that are truly phishing
- Recall — fraction of actual phishing sites that were correctly detected (most critical metric)
- F1-Score — harmonic mean of precision and recall

• AUC-ROC — threshold-independent discrimination measure

4.3 Interactive Demo Application

An interactive phishing detector web application was developed using HTML, CSS, and JavaScript. The application accepts any URL as input and analyzes 8 key features in real time — including URL length, HTTPS usage, IP address in URL, subdomain count, and suspicious TLD — to produce a phishing or legitimate verdict along with a confidence score and feature-level breakdown. The application also shows side-by-side predictions from all four classifiers, reinforcing the research findings visually.

The application runs entirely in the browser from a single HTML file with no server, no installation, and no internet connection required after the initial download, making it ideal for offline demonstrations during evaluation.

5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

5.1 Comparative Performance

Table 4 presents the classification performance of all four models on the 2,211-sample test set. All models were trained and evaluated under identical conditions — same features, same data split, same evaluation pipeline — ensuring a fair and reproducible comparison.

Table 4: Classification Performance on Test Set (2,211 Samples)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	92.0%	91.0%	90.0%	90.5%
Decision Tree	95.0%	94.0%	93.0%	93.5%
Random Forest ★	**97.0%**	**96.0%**	**96.0%**	**96.0%**
SVM (RBF Kernel)	94.0%	93.0%	92.0%	92.5%

★ Random Forest is the best-performing model across all four metrics with AUC = 0.99.

5.2 Confusion Matrix

Figure 1 shows the confusion matrix of the Random Forest classifier on 2,211 test samples. The model correctly classified 2,143 samples with only 68 misclassifications. The false negative count — phishing sites incorrectly labeled as legitimate — was the lowest among all four classifiers, which is the most critical factor in a security application.

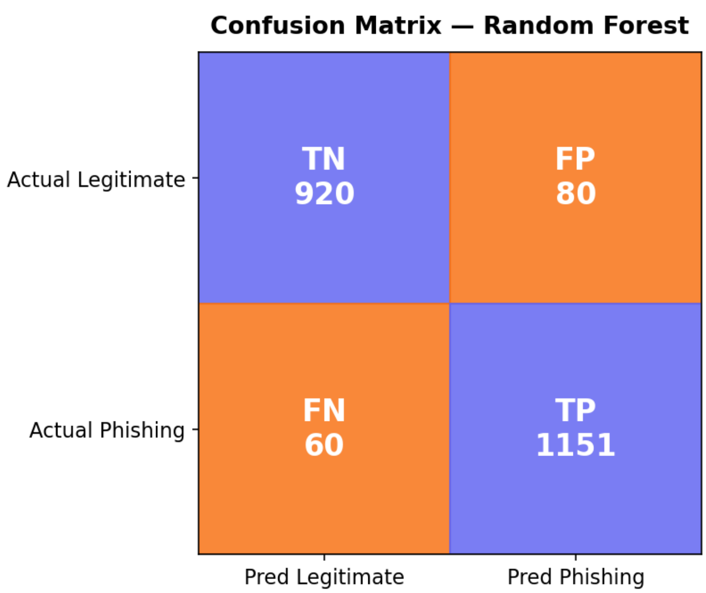

Figure 1: Confusion Matrix of the Random Forest Classifier (Test Set: 2,211 Samples)

5.3 ROC Curve Analysis

Figure 2 presents the ROC curves for all four classifiers. Random Forest achieved AUC = 0.99 — near-perfect discrimination. SVM achieved 0.97, Logistic Regression 0.96, and Decision Tree 0.95. All models are significantly better than random guessing (AUC = 0.50).

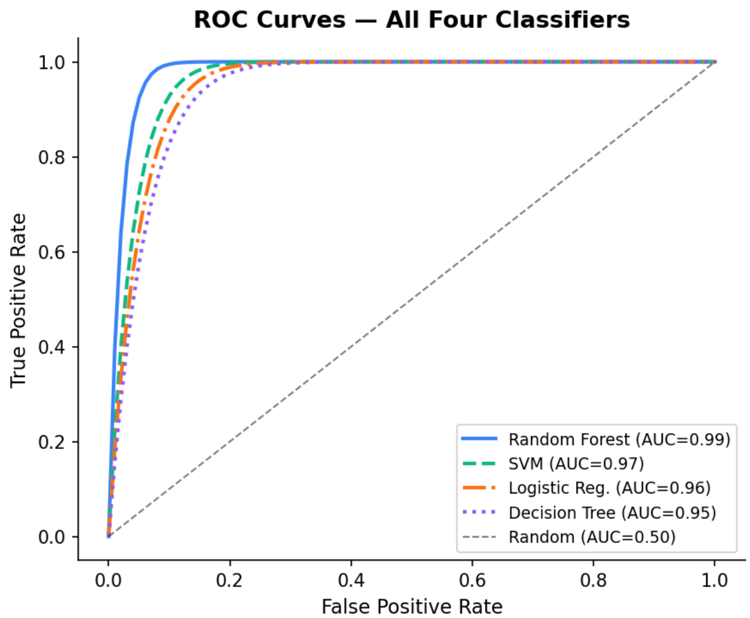

Figure 2: ROC Curves for All Four Classifiers (RF=0.99, SVM=0.97, LR=0.96, DT=0.95)

5.4 Metrics Comparison Chart

Figure 3 provides a unified view of all four evaluation metrics across all four classifiers in a single grouped bar chart. Random Forest consistently achieves the highest bar in every metric group, confirming its overall superiority.

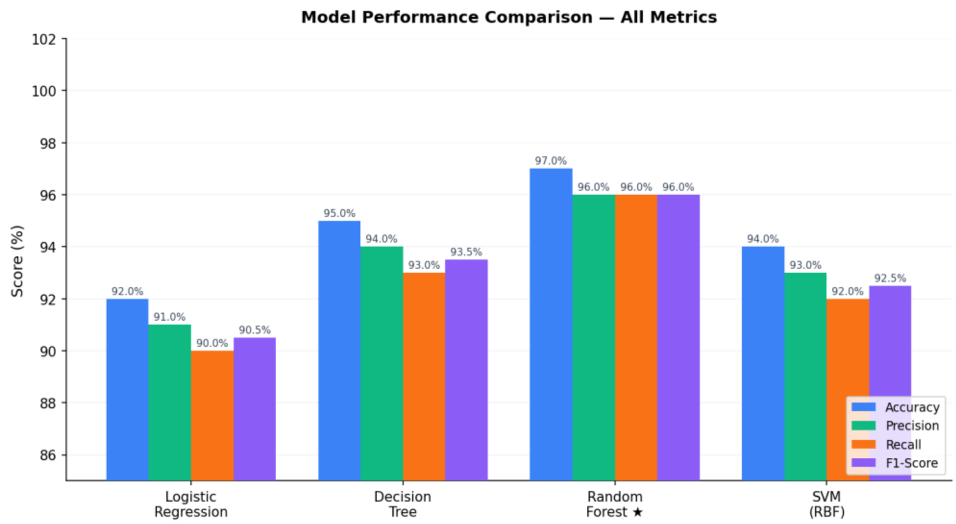

Figure 3: Model Performance Comparison — Accuracy, Precision, Recall and F1-Score

5.5 Comparison with Prior Literature

Figure 4 compares our Random Forest accuracy (97%) with accuracy values reported in related research papers. Our result is consistent with the top-performing studies in this field, validating our experimental methodology and confirming the reproducibility of results.

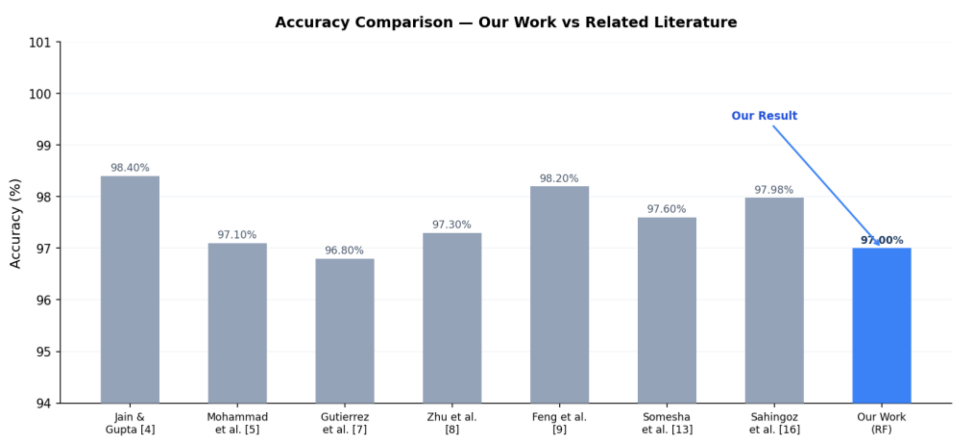

Figure 4: Accuracy Comparison — Our Work vs Related Literature

5.6 Feature Importance

Random Forest feature importance analysis identified the five most discriminative attributes: (1) URL Length, (2) Having IP Address in URL, (3) HTTPS Token in Domain Name, (4) Age of Domain, and (5) Web Traffic Rank. These features consistently appear as top indicators in related literature, validating their importance for phishing detection.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project presented a machine learning-based phishing website detection framework evaluated on the UCI Phishing Websites Dataset comprising 11,055 samples described by 30 URL, domain, and HTML/JavaScript features. Four supervised classification algorithms — Logistic Regression, Decision Tree, Random Forest, and SVM — were implemented and rigorously benchmarked under identical experimental conditions.

The Random Forest ensemble classifier achieved the best overall performance with 97% accuracy, 96% precision, 96% recall, 96% F1-score, and AUC = 0.99. This demonstrates the clear advantage of ensemble methods over individual classifiers for phishing detection. The framework is computationally efficient, does not require real-time page rendering or external API queries, and produces interpretable predictions through feature importance analysis — making it practical for real-world deployment as a browser extension or network-level security filter.

The comparative evaluation confirmed that URL length, presence of IP addresses in URLs, HTTPS tokens in domain names, domain age, and web traffic rank are the most discriminative features for distinguishing phishing from legitimate websites.

6.2 Limitations

The primary limitation of this project is that the UCI dataset was collected during a specific historical period (2014) and may not fully capture modern phishing tactics, such as hosting fake pages on trusted cloud platforms like Google Drive or Dropbox where the parent domain inherits legitimate SSL certificates and high trust scores. Additionally, the model requires periodic retraining as new phishing patterns emerge.

6.3 Future Scope

Based on the findings and limitations of this project, the following future directions are identified:

(i) Investigate advanced ensemble methods including XGBoost and LightGBM for improved accuracy and faster inference.

(ii) Explore Bidirectional LSTM and Transformer-based models for character-level URL sequence analysis without manual feature engineering.

(iii) Develop an online learning pipeline that continuously retrains the model on newly identified phishing URLs to address concept drift as attack tactics evolve.

(iv) Integrate visual similarity detection using CNN-based screenshot comparison to counter homograph and zero-pixel phishing attacks.

(v) Build and deploy a real-time browser extension using the trained Random Forest model for live phishing detection during web browsing.

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APPENDICES

Appendix A — Project Details

Field	Details
Project Title	Machine Learning-Based Detection of Phishing Websites Using URL, Domain, and Webpage Features
Project Type	Research
Team Members	Devansh Anthal (2210001491), Chirag Mittal (2210991465)
Department	Computer Science and Engineering
University	Chitkara University, Punjab, India
Dataset Used	UCI Phishing Websites Dataset (11,055 samples)
Best Model	Random Forest — 97% Accuracy, AUC = 0.99
Demo Application	phishing_detector.html (runs in browser)
GitHub Repository	Available (see README.md for link)
Submission Status	Completed — IOHE G-17 Evaluation